

Abhishek Aylaveni

Embedded Systems / Firmware / IoT / Edge AI

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Portfolio

Professional Profile

Master's student in Electronics and Embedded Systems at ESIGELEC with practical experience in embedded firmware, IoT and hardware validation. Skilled in Embedded C, STM32, nRF52840 and Zephyr, with hands-on work in sensor integration, real-time monitoring and TinyML evaluation. Combines systematic debugging with reproducible testing and clear technical documentation.

Experience

Research Intern

KU Leuven, Bruges — *Oct. 2026 – Present*

- Implementing and validating MLPerf Tiny benchmarks on nRF52840 DK with Zephyr and TensorFlow Lite Micro.
- Validated 48,608 anomaly-detection windows with exact host agreement and reproducible builds.
- Planned outcomes: assess bit-flip reliability through fault injection, evaluate software fault tolerance, and document comparative results in a final technical report.

Embedded Systems Intern

Emertxe Information Technologies — *Jun. – Jul. 2026*

- Developed a real-time EV dashboard and ADAS warning system using STM32F103C8T6 and PICSimLab.
- Integrated ultrasonic sensing, collision and blind-spot warnings, vehicle state management and UART telemetry.

Electrical Trainee Engineer

Indian Power Projects Ltd — *Dec. 2022 – Nov. 2023*

- Supported transformer and control-panel design, testing and preventive maintenance.
- Contributed to a 10% reduction in unplanned downtime and assisted real-time SCADA monitoring.

Projects

LOCAT'IT LoRa Prototype

STM32, LoRa — *Oct. 2025 – Dec. 2025*

- Contributed hardware integration and testing to a team LoRa prototype with STM32, SX1278 and an OLED receiver.

Autonomous Line-Following Robot

ARM Cortex-M, Embedded C — *Apr. 2025 – Jun. 2025*

- Implemented PID trajectory control with 90% accuracy; reduced processing latency by 20%.

Smart Parking IoT System

ESP / Wi-Fi, Ultrasonic Sensors — *Apr. 2022 – Jul. 2022*

- Built an ultrasonic and Wi-Fi parking dashboard; reduced estimated search time by 30%.

Education

M.Sc. Electronics and Embedded Systems

ESIGELEC, Rouen, France — *2025 – 2026*

Real-time systems, microcontrollers, sensors, wireless communication and validation.

Bachelor Electrical and Electronics Engineering

TKR Engineering College, India — *2018 – 2022*

Electronics, electrical systems, embedded projects, and IoT.

Technical Skills

Languages: Embedded C, C, Python, test scripts
Embedded: STM32, ESP32, nRF52840, ARM Cortex-M, Arduino, Zephyr, RTOS / FreeRTOS, TinyML
Protocols: LoRaWAN, MQTT, CAN, UART, SPI, I2C, Wi-Fi
Validation: Oscilloscope, logic analyzer, hardware debugging, test plans, functional testing, prototype validation
Tools: Git, GitHub, VS Code, Linux, MATLAB, Simulink, PICSimLab, TensorFlow Lite Micro

Languages

English – C2
French – Intermediate

Professional Strengths

- Reading datasheets, electronic schematics, and technical documentation.
- Functional testing, hardware debugging, and prototype validation.
- Technical English communication and collaboration in multicultural engineering teams.